DATABASES

Organizing, storing, and managing the world's data

A foundational guide to concepts, models, and everyday use



What Is a Database?

A database is an organized collection of structured information, stored electronically so it can be easily accessed, managed, and updated.

- Stores data in a structured, searchable way
- Controls who can view or change information
- Keeps data consistent as it's added or updated
- Lets users find exactly what they need, fast

customers

id	name	email	city
1	Ama Owusu	ama@mail.com	Accra
2	Kojo Mensah	kojo@mail.com	Kumasi
3	Efua Asante	efua@mail.com	Tema

Rows = records • Columns = fields

Database vs. Spreadsheet

Why not just use Excel?

Spreadsheet

- Good for small, simple data
- One user editing at a time works best
- Little protection against duplicate or bad data
- Struggles past a few thousand rows

Database

- Built for large, complex, connected data
- Many users can read and write at once, safely
- Enforces rules so data stays accurate
- Scales to millions or billions of records

Two Broad Families



Relational (SQL)

Data lives in tables with fixed columns, linked by keys. Strong consistency and mature tooling.

EXAMPLES

MySQL, PostgreSQL, Oracle, SQL Server



Non-Relational (NoSQL)

Flexible structures — documents, key-value pairs, graphs. Built for scale and changing data shapes.

EXAMPLES

MongoDB, Redis, Cassandra, Neo4j

The DBMS: Software in Charge

A Database Management System (DBMS) is the software layer that sits between users and raw data files — handling storage, retrieval, security, and updates.



Storage & Retrieval

Reads and writes data efficiently to disk or memory



Security

Controls permissions and protects sensitive data



Concurrency

Lets many users work at once without conflicts



Integrity

Enforces rules so data stays accurate and valid

Popular DBMS platforms:

MySQL | PostgreSQL | Microsoft SQL Server | Oracle | MongoDB | SQLite

The Relational Model

Data is split into related tables, connected through shared key values instead of being duplicated everywhere.

orders

order_id	customer_id
101	1
102	2
103	1

foreign key → primary key

customers

id	name
1	Ama Owusu
2	Kojo Mensah
3	Efua Asante

Each order references a customer by ID — no need to repeat the customer's full details in every order row.

Keys: Connecting the Data



Primary Key

Uniquely identifies each row in a table — no two rows share one.



Foreign Key

A field that points to a primary key in another table, creating a relationship.



Composite Key

Two or more columns combined to uniquely identify a row.

Example: in the customers table, id is the primary key. In orders, customer_id is a foreign key pointing back to it.

SQL: Talking to the Database

SQL (Structured Query Language) is how you ask a relational database for what you want.

CREATE / INSERT	Add new tables or new rows of data
SELECT	Retrieve and filter existing data
UPDATE	Modify data that's already stored
DELETE	Remove rows that are no longer needed

```
SELECT name, city
FROM customers
WHERE city = 'Accra'
ORDER BY name ASC;
```

Plain-English translation:

"Show me the name and city of every customer who lives in Accra, sorted alphabetically."

Normalization: Cutting Out Redundancy

A step-by-step process for organizing tables so each fact is stored in exactly one place.

1NF

Atomic Values

Each cell holds a single value — no lists or repeating groups packed into one field.

2NF

Full Dependency

Every non-key column depends on the whole primary key, not just part of it.

3NF

No Transitive Links

Non-key columns depend only on the key — not on each other.

ACID: Trustworthy Transactions

Four guarantees that keep data reliable, even when things go wrong mid-operation.



A

Atomicity

A transaction either fully completes or doesn't happen at all.



C

Consistency

Every transaction moves the database from one valid state to another.



I

Isolation

Concurrent transactions don't interfere with each other.



D

Durability

Once committed, data survives even a crash or power loss.

Inside NoSQL: Four Flavors



Document

JSON-like documents grouped in collections

MongoDB, Couchbase



Key-Value

Simple pairs, extremely fast lookups

Redis, DynamoDB



Column-Family

Columns grouped for fast analytics at scale

Cassandra, HBase



Graph

Nodes and edges model relationships directly

Neo4j, ArangoDB

Databases in Everyday Life



Banking

Tracks balances and processes millions of transactions safely each day



E-commerce

Manages product catalogs, orders, and inventory in real time



Social Media

Stores billions of posts, profiles, and connections between users



Healthcare

Keeps patient records accurate, secure, and accessible to care teams

Key Takeaways

1. A database organizes information so it's easy to store, find, and trust
2. A DBMS is the software that manages access, security, and integrity
3. Relational databases use linked tables; NoSQL trades some structure for flexibility and scale
4. Keys, normalization, and ACID guarantees are what keep data clean and reliable

